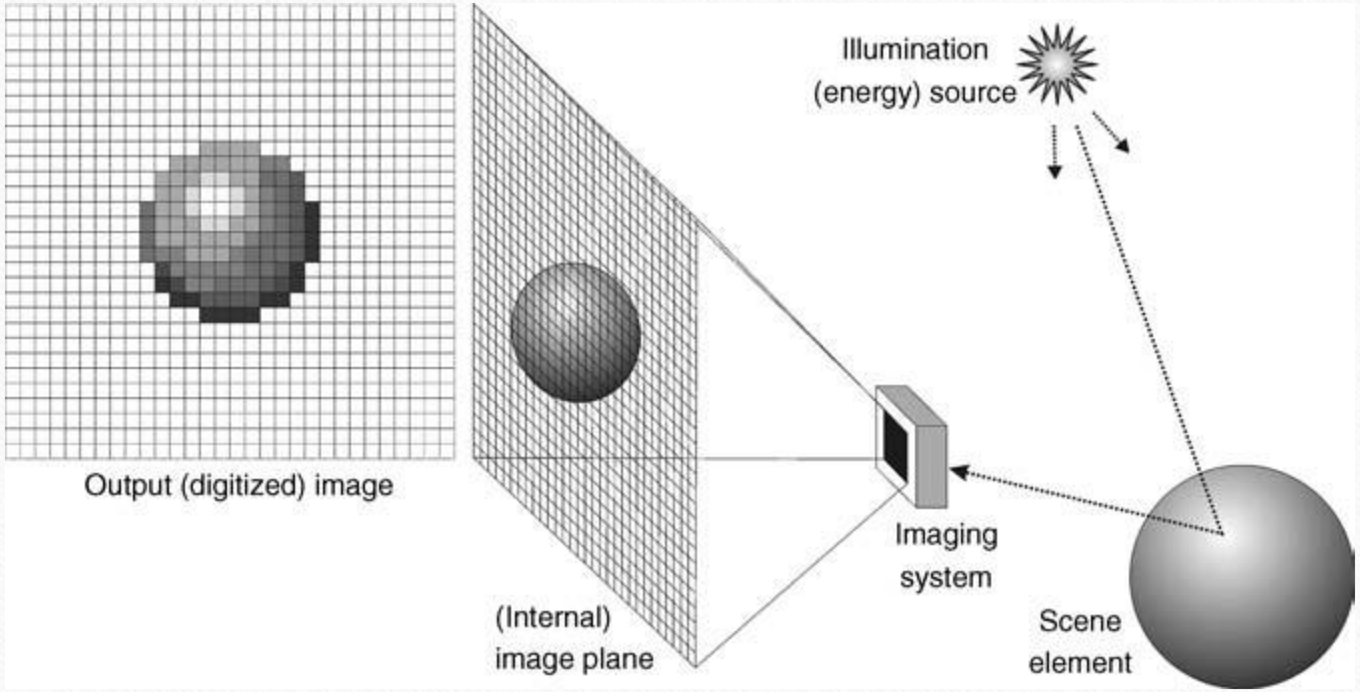
CHAPTER 5

IMAGE SENSING AND ACQUISITION

WHAT WILL WE LEARN?

- What are the main parameters involved in the design of an image acquisition solution?
- How do contemporary image sensors work?
- What is image digitization and what are the main parameters that impact the digitization of an image or video clip?
- What is sampling?
- What is quantization?
- How can I use MATLAB to resample or requantize an image?

Image acquisition, formation, and digitization



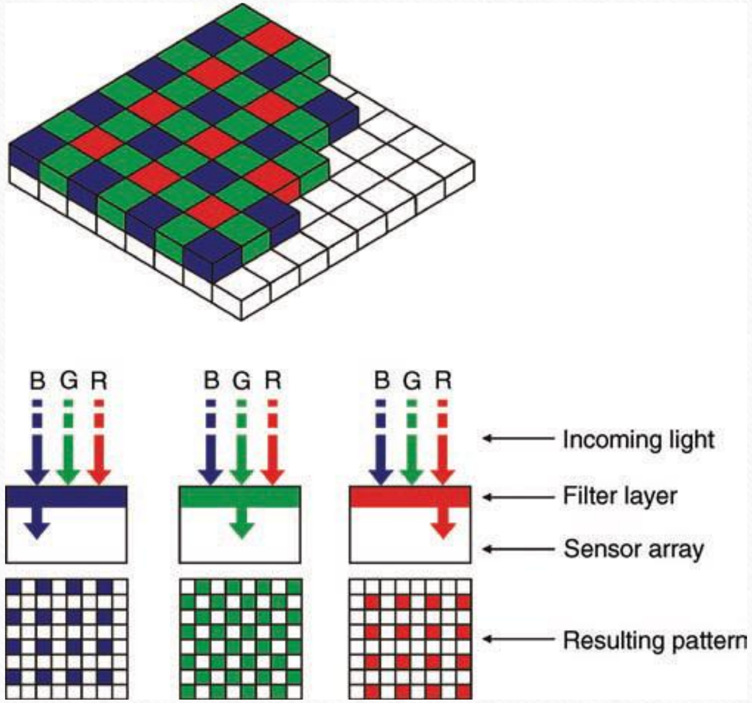
Types of Images

- Images can be classified into three categories
  - *Reflection Images (ảnh phản chiếu* - surfaces of objects)
  - *Emission Images (ảnh phát xạ - stars and light bulbs)*
  - *Absorption Images (ảnh hấp thụ) (X-ray image)*

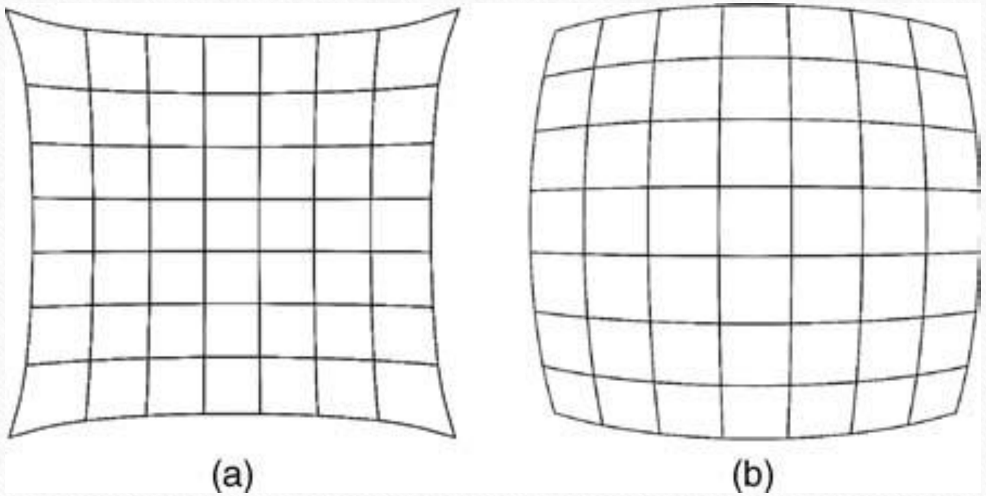
Color Encoding and Representation

- Human perception of light is describbed in 3 parameters:
  - *Brightness*
  - *Hue*
  - *Saturation*

IMAGE ACQUISITION



Camera Optics



Examples of lens aberrations:

(a) pincushion distortion; (b) barrel distortion

IMAGE DIGITIZATION

- Two processes
  - *Sampling*
  - *Quantization*

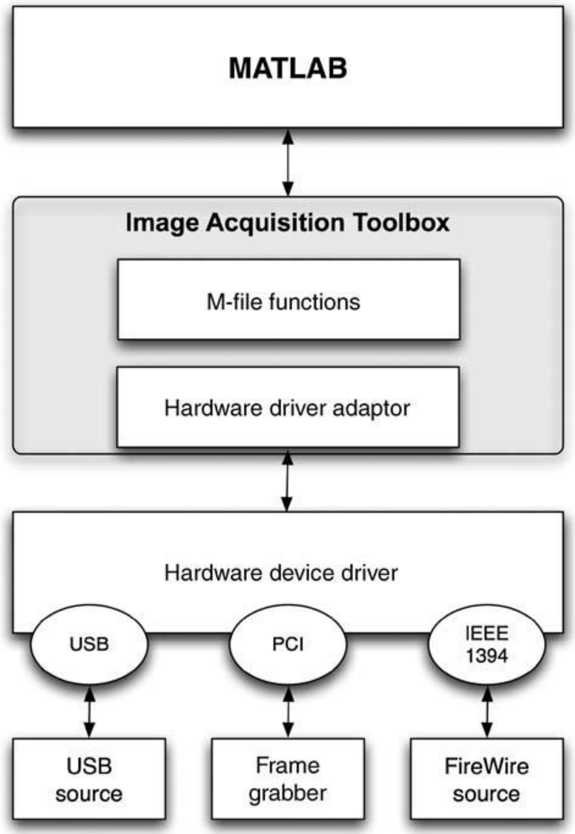
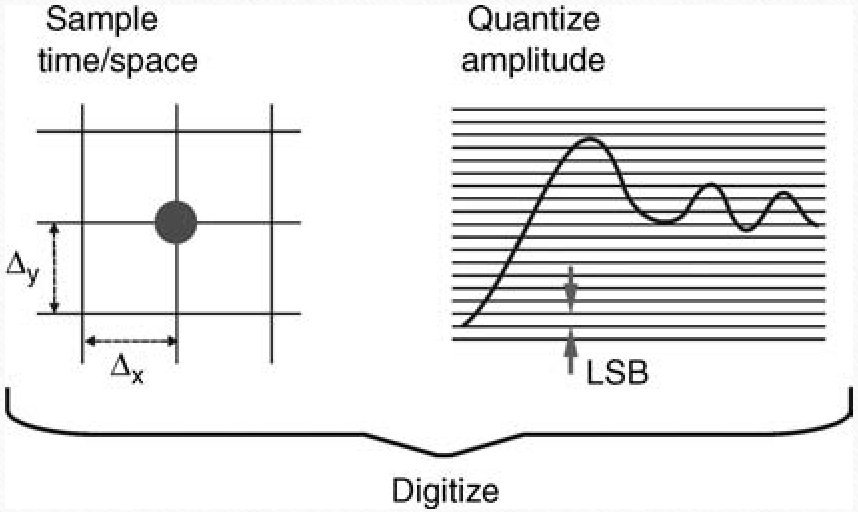


IMAGE DIGITIZATION

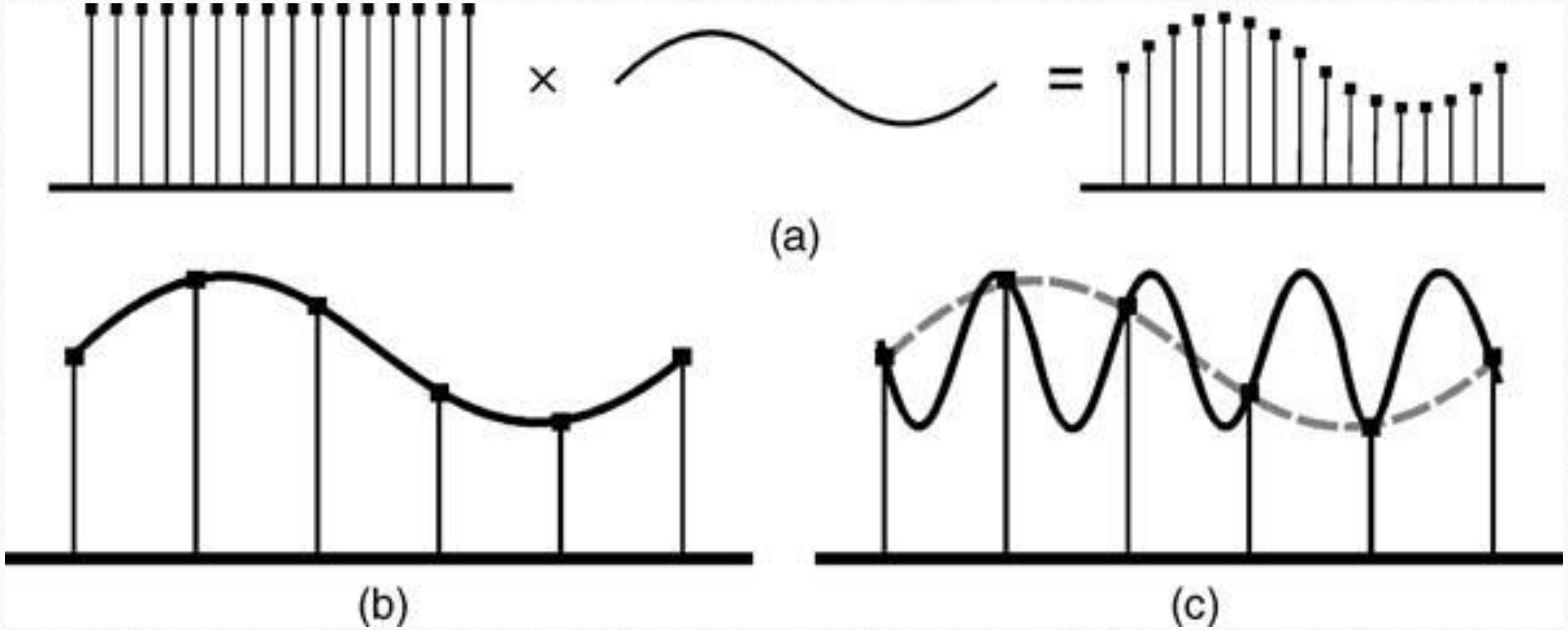


Digitization = sampling + quantization

Sampling

- Sampling is the process of measuring the value of a 2D function at discrete intervals along the x and y dimensions.
- Two parameters must be taken into account when sampling images
  - *sampling rate:* number of samples across the height and width of the image
  - *sampling pattern:* the physical arrangement of the samples

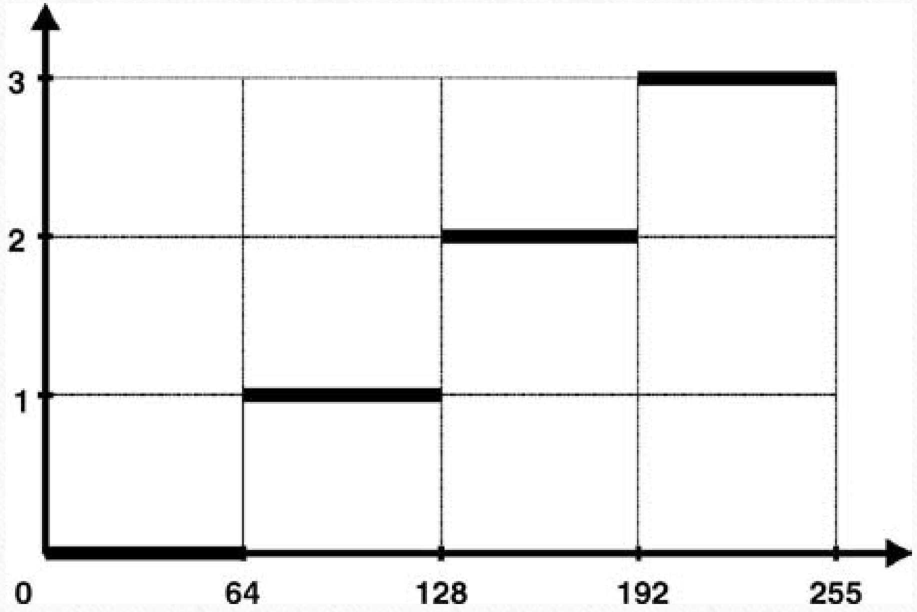
Sampling



(a) sampling process, (b) result of reconstructing a signal from an appropriate number of samples,
(c) the same number of samples as in (b) would be insufficient to reconstruct a signal with higher frequency.

Quantization

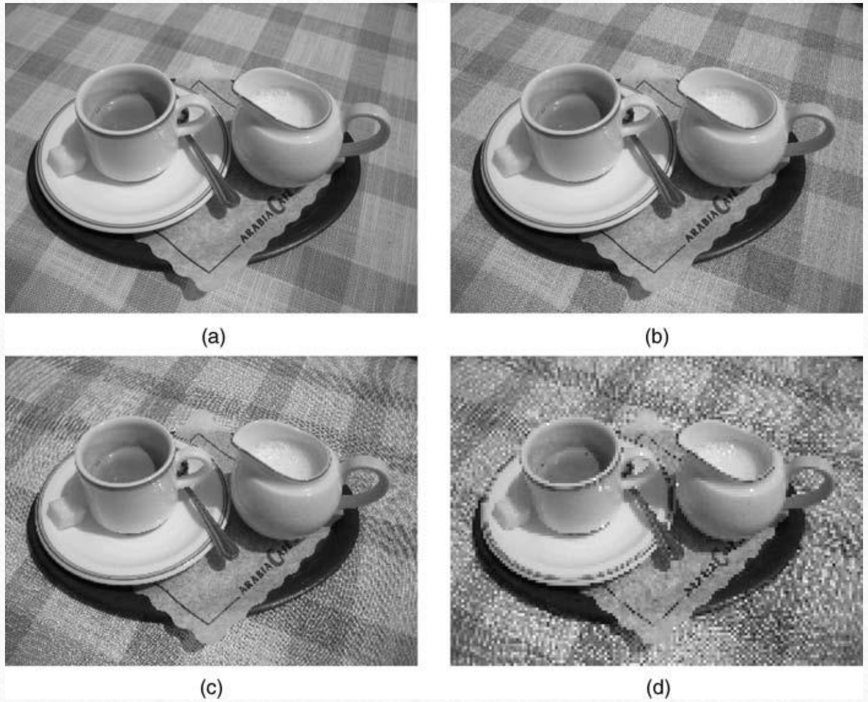
- Quantization is the process of replacing a continuously varying function with a discrete set of quantization levels.
- Image quantization can be described as a mapping process by which groups of data points (several pixels within a range of gray values) are mapped to a single point.



A mapping function for uniform quantization ($N = 4$).

Spatial and Gray-Level Resolution

- Spatial resolution expresses the density of pixels in an image (dots per inch (dpi))
- Example: the original image (a) is of size 1944 × 2592 and it is displayed at 1250 dpi. The three other images (b - d) have reduced spatial resolution to 300, 150, and 72 dpi, respectively



In MATLAB

- **`grayslice`**

```
I1 =  imread('ml_gray_640_by_480_256.png');
I2 = grayslice(I1,128); figure, imshow(I2,gray(128));
I3 = grayslice(I1,64); figure, imshow(I3,gray(64));
```

PROBLEMS

- 5.1-5.3 page 101-102 in textbook
- Practice: write a function to quantize an image.

  (see Quantization.m)